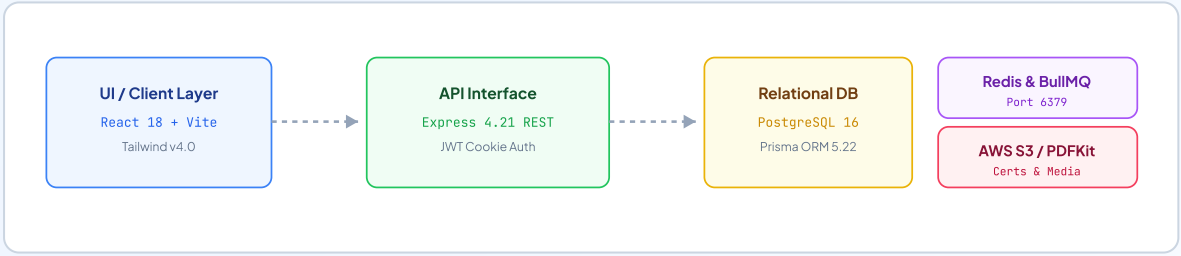
FORGE LOOM LMS

IEEE 830 Standard Compliance

Software Requirements Specification (SRS)

Formal engineering specification defining functional requirements, external system interfaces, non-functional quality attributes, and data constraints for the FORGE Loom multi-tenant enterprise LMS.



SPECIFICATION STANDARD
IEEE Std 830–1998 Conformance

SYSTEM TARGET
50,000 Concurrent Active Users

SECURITY COMPLIANCE
Zero-Trust Multi-Tenant Partitioning

BACKEND ENDPOINTS
22 Active Routers (Express 4.21)

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1 System Overview & Scope

The **FORGE Loom LMS** is a distributed, web-based Learning Management System designed to orchestrate educational tracks, innovation incubator sprints, mentor advisories, and career placements across higher-education institutions and enterprise partners.

Operating Environment & Hardware Interfaces

Environment

- **Server OS:** Linux (Ubuntu 22.04 LTS / Alpine containerized) or macOS Darwin.
- **Runtime:** Node.js v20.x or v22.x LTS (ESM modules).
- **Database Engines:** PostgreSQL 15+ (Port 5433) and Redis 7+ (Port 6379).
- **Client Support:** Modern evergreen web browsers (Chromium 115+, Firefox 115+, Safari 16+). Responsive from 360px mobile to 4K ultra-wide monitors.

Core Functional Requirements (IEEE Functional Breakdown)

Req ID	Feature Name	Requirement Specification	Priority
FR-01	Authentication & JWT Transport	The system shall authenticate users via email and bcrypt-hashed password, issuing a signed JWT stored inside an HTTP-only, secure cookie.	Critical
FR-02	College Tenant Scoping	The system shall enforce row-level scoping via `collegeId` for students, mentors, trainers, and college administrators, preventing cross-tenant reads or writes.	Critical
FR-03	Video Timestamp Sync	The client shall persist video playback progress every 5 seconds via `POST /api/video-progress`, resuming automatically on subsequent sessions.	High
FR-04	Citadel Sprint State Machine	The system shall transition student team sprints through states: `not_started → in_progress → submitted → reviewed → approved`. Sprint 3 approval unlocks Investor Access.	Critical
FR-05	1-on-1 Mentor Booking	The system shall allow students to query available mentor slots and submit bookings. Mentors shall accept, reschedule, or cancel bookings via `/api/bookings`.	High
FR-06	Asynchronous Score Queue	The scoring service shall dispatch recalculation jobs to Redis/BullMQ whenever attendance, milestones, or evaluations change, computing: `(Events*0.2) + (Projects*0.4) + (Mentor*0.3) + (Team*0.1)`.	Critical
FR-07	Verifiable Dynamic Certs	Upon course completion, the server shall generate a PDF certificate with embedded QR code. Scanning the QR shall resolve public endpoint `/verify/:token` without authentication.	High
FR-08	HR Talent Pool Search	The system shall provide full-text search and multi-facet filtering (domain tags, skill sets, Builder Score slider) across pre-screened student profiles.	High

2 Non-Functional Requirements (NFRs)

The platform enforces strict performance, reliability, security, and maintainability benchmarks:

Performance & Throughput

Latency

- **API Latency:** 95th percentile response time < 50ms for cached reads; < 150ms for transactional writes.
- **Concurrent Capacity:** Architected to support 5,000 to 50,000 concurrent active users via stateless horizontal API replication.
- **Asynchronous Processing:** BullMQ queue workers process certificate generation and score recomputations within 3 seconds of trigger.

Security & Data Integrity

Hardening

- **Cryptographic Password Storage:** Passwords hashed using bcrypt with salt rounds ≥ 10.
- **HTTP Hardening:** Helmet middleware sets Content-Security-Policy, X-Frame-Options, and Strict-Transport-Security headers.
- **SQL Injection Immunity:** Prisma ORM utilizes parameterized prepared queries exclusively.

Reliability & Fault Tolerance

Uptime

- **Service Availability:** Designed for 99.9% uptime with health check endpoints at `GET /health`.
- **Database Resilience:** PostgreSQL foreign key cascading rules and unique constraints maintain referential integrity.
- **Graceful Shutdown:** Server traps `SIGTERM` / `SIGINT` to close Prisma pools and BullMQ connections cleanly.

Maintainability & Code Hygiene

Standards

- **Strict TypeScript:** Zero implicit `any`; strict typecheck passes across all three workspaces.
- **Shared Domain:** Zero model divergence between frontend React views and backend Prisma database schemas.
- **Clean Repository:** Git repository strictly excludes database dumps, logs, and secret files.

Engineering Compliance Sign-off

This Software Requirements Specification represents the verified implementation in the official repository: `github.com/Winnovationforge/Forge-loom`.